Expectation for $g(X, Y)$

▽ Let $Z = g(X, Y) : R^2 \to R$

$$
= \begin{cases} X \\ Y \\ X + Y \\ X \cdot Y \end{cases}
$$

$\min(X, Y)$ ...

▽ $E(Z) = E(g(X, Y))$

$$
= \begin{cases} \sum\sum_{\forall (x,y)} g(x, y) \cdot P_{X,Y}(x, y) \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) \cdot f_{X,Y}(x, y) \end{cases}
$$

Ex. Let (X, Y) have joint PDF:

$$f_{X,Y}(x, y) = \begin{cases} 1.2(x^2 + y), & 0 < x < 1 \\ & 0 < y < 1 \\ 0, & \text{o.w.} \end{cases}$$

Find $E(X + Y)$.

Approach ① $E(X+Y) = E(X) + E(Y)$

$$E(X) = \int\int x \cdot f_{X,Y}(x, y)\, dy\, dx$$

$$E(Y) = \int\int y \cdot f_{X,Y}(x, y)\, dx\, dy$$

$\therefore$ Two "$\int\int$" to solve.

Approach ② $E(X+Y) = E(g(X,Y))$

$$E(X+Y) = \int_0^1 \int_0^1 (x+y) \cdot 1.2 \cdot (x^2+y)\, dx\, dy$$

$$= 1.2 \left(\int_0^1 \int_0^1 x^3 + xy + yx^2 + y^2 \, dx\, dy \right)$$

$$= 1.2 \cdot \left(\int_0^1 \frac{1}{4} + y \cdot \frac{1}{2} + y \cdot \frac{1}{3} + y^2 \, dy \right)$$

$$= 1.2 \cdot \left(\frac{1}{4} + \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \right)$$

$$= 1.2$$

Ex. Let (X, Y) have joint PDF:

$$f_{X,Y}(x, y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{o.w.} \end{cases}$$

Find $E(X \cdot Y^2)$.

$$E(X \cdot Y^2) = \int_0^1 \int_0^y \underbrace{x\, y^2}_{g(x,y)} \cdot 2 \, dx \, dy$$

$$= 2 \cdot \left(\int_0^1 y^2 \cdot \frac{1}{2} \cdot y^2 \, dy \right)$$

$$= \frac{1}{5}$$